INCH-POUND

MIL-DTL-46077H
17 January 2025
SUPERSEDING
MIL-DTL-46077G
w/AMENDMENT 1
26 January 2017

DETAIL SPECIFICATION

ARMOR PLATE, TITANIUM ALLOY, WELDABLE

This specification is approved for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers weldable wrought titanium alloy structural armor plate in nominal thicknesses from 0.125 - 4.0 inch, inclusive.

1.2 Classification. Wrought titanium alloy armor should be of the following classes as specified (see 6.2).

1.2.1 Class 1. Class 1 is equivalent to material specified by previous revisions of MIL-A-46077, which is an Extra Low Interstitial (ELI) grade (see table I).

1.2.2 Class 2. Class 2 is an alpha-beta titanium alloy, specifically, 6Al-4V with intermediate oxygen levels ranging from greater than 0.14 to 0.20 weight percent (see table I and 6.1).

1.2.3 Class 3. Class 3 permits oxygen levels beyond 0.20 weight percent suitable for applications such as appliqué where maximum ductility and toughness are not necessary (see table I).

Comments, suggestions, or questions on this document should be addressed to: Director, U.S. Army DEVCOM, Army Research Laboratory, Army Research Directorate, Specifications and Standards Office, Attn: FCDD-RLA-M, Aberdeen Proving Ground, MD 21005-5069. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil/.

AMSC N/A FSC 9535

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1.2.4 Class 4. Class 4 alloy compositions consist of Al, V, & Fe chemistries outside the conventional Ti-6Al-4V chemistries.

1.2.4.1 Class 4a. Class 4a specific chemistry requirements are listed in table I.

1.2.4.2 Class 4b. Class 4b specific chemistry requirements are listed in table I.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-129	-	Military Marking for Shipment and Storage
MIL-STD-662	-	V_{50} Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

American Society for Quality Control (ASQC)

ASQ Z1.4	-	Procedures, Sampling and Tables for Inspection by Attributes

(Copies of this document are available online at https://asq.org.)

ASTM INTERNATIONAL

ASTM A480/A480M	-	Standard Specification for General Requirements for Flat-Rolled Stainless and Heat-Resisting Steel Plate, Sheet, and Strip
ASTM E8/E8M	-	Standard Test Methods for Tension Testing of Metallic Materials

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ASTM E539	-	Standard Test Method for X-Ray Emission Spectrometric Analysis of 6Al-4V Titanium Alloy
ASTM E1409	-	Standard Test Method for Determination of Oxygen and Nitrogen in Titanium and Titanium Alloys by the Inert Gas Fusion Technique
ASTM E1447	-	Standard Test Method for Determination of Hydrogen in Titanium and Titanium Alloys by Inert Gas Fusion Thermal Conductivity/Infrared Detection Method
ASTM E2371	-	Standard Test Method for Analysis of Titanium and Titanium Alloys by Direct Current Plasma and Inductively Coupled Plasma Atomic Emission Spectrometry (Performance-Based Test Methodology)

(Copies of these documents are available online at https://www.astm.org/.)

SAE INTERNATIONAL

AMS 2249	-	Chemical Check Analysis Limits - Titanium and Titanium Alloys
AMS 4903	-	Titanium Alloy Sheet, Strip, and Plate, 6Al - 4V, Solution Heat Treated
AMS 4904	-	Titanium Alloy Sheet, Strip, and Plate 6Al - 4V, Solution Heat Treated and Aged
AMS 4911	-	Titanium Alloy, Sheet, Strip, and Plate, 6Al - 4V, Annealed
ARP 982	-	Minimizing Stress Corrosion Cracking in Wrought Titanium Alloy Products

(Copies of these documents are available online at https://www.sae.org/.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the contractor shall furnish a sample or samples for first article inspection and approval. The contractor shall comply with this requirement at the time of his first order or contract and at any time that the supplier has not furnished wrought armor in the applicable thickness range under this specification within a period of 18 months (see 4.4).

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3.2 Chemical composition.

3.2.1 Class compositions. The chemical composition of the plates shall be as specified in table I for classes 1, 2, 3, 4a, and 4b. If no class is specified on the contract, drawing or order, class 1 shall be supplied. Chemical composition variances shall meet the requirements of SAE AMS 2249.

3.2.1.1 Class 4.

3.2.1.1.1 Class 4 composition. Upon approval of the procuring activity, an alternate composition of titanium alloy may be authorized. In addition to the composition limits specified in table I, suppliers must submit to the procuring activity established limits for each intentionally added element as well as maximum limits for others.

3.2.1.1.2 Class 4 density. The density (see 6.9) shall be 0.158 through 0.165 pounds per cubic inch (4.373 g/cm^3 through 4.567 g/cm^3). Density may be calculated based on chemical composition range established.

3.2.1.2 Class 4a. This specific composition of a class 4 titanium alloy has been tested and approved as an alternative composition of a class 4 titanium alloy.

3.2.1.3 Class 4b. This specific composition of a class 4 titanium alloy has been tested and approved as an alternative composition of a class 4 titanium alloy.

TABLE I. Chemical composition - percent by weight. 1/

Class	Al	V	C	O	N	H 2/	Fe	Ti 3/	Other 4/	
									Each	Total
1	5.50 - 6.50	3.50 - 4.50	0.04	0.14	0.02	0.0125	0.25	Rem	0.10	0.40
2	5.50 - 6.75	3.50 - 4.50	0.08	0.20	0.05	0.0150	0.30	Rem	0.10	0.40
3	5.50 - 6.75	3.50 - 4.50	0.08	0.27	0.05	0.0150	0.30	Rem	0.10	0.40
4	5/	5/	0.08	0.27	0.05	0.0150	5/	Rem	5/	5/
4a	3.5 - 4.5	2.0 - 3.0	0.08	0.27	0.03	0.0150	1.2 - 1.8	Rem	0.10	0.30
4b	4.2 - 5.4	2.5 - 3.5	0.08	0.15 - 0.19	0.03	0.0150	0.5 - 0.7	Rem	0.10	0.30

1/ Where single units are shown, these indicate the maximum amounts permitted.
2/ Hydrogen shall be determined on each lot of the product as shipped.
3/ Titanium is determined by differences.
4/ Other elements need not be analyzed nor reported unless otherwise specified.
5/ For class 4 alternate composition details see 1.2.4 and 3.2.1.1.1.

3.3 Mechanical properties. Plate shall meet the minimum mechanical properties of table II in both the longitudinal and transverse directions after final heat treatment (See 4.6.1.2 and 4.6.2.2). After temperature exposure and cooling, a minimum of 0.001 inch shall be removed from all surfaces of the sample prior to testing. Sufficient material shall be removed so as to ensure compliance of the sample with the requirements of 3.9.

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TABLE II. Minimum mechanical properties.

Class	Thickness	Yield Strength 0.2% Offset (psi)	Tensile Strength (psi)	Elongation (%)
1	0.125" to 4.0"	110,000	120,000	10
2	0.125" to 1.000"	120,000	130,000	10
2	1.001" to 2.000"	110,000	125,000	10
2	2.001" to 4.000"	110,000	125,000	8
3	0.125" to 4.0"	110,000	120,000	6
4	0.125" to 4.0"	110,000	120,000	6
4a	0.125" to 1.000"	115,000	130,000	10
4a	1.001" to 2.000"	110,000	125,000	10
4a	2.001" to 4.000"	110,000	125,000	8
4b	0.125" to 1.000"	110,000	120,000	10
4b	1.001" to 2.000"	110,000	120,000	10
4b	2.001" to 4.000"	110,000	120,000	10

3.4 Ballistic requirements. The protection ballistic limit, BL(P), shall be as specified in Appendix A. No through cracks greater than 2.5 inch in length on any plate surface are permitted after ballistic testing. When a complete penetration cannot be obtained for any class of armor material, the following rule shall be in effect until a new ballistic acceptance round can be developed and utilized. When the ballistic velocities of four (4) partial penetrations are above the minimum ballistic requirement for the specific thickness, the material shall be certified as acceptable with a V_{50} (which obviously cannot be explicitly determined) above the minimum requirement.

3.5 Dimensional tolerances.

3.5.1 Length, width, and thickness. Length and width tolerances shall be as agreed upon by purchaser and vendor. The plus and minus tolerances on plate thickness shall be as specified in table III.

TABLE III. Plate thickness tolerances.

Thickness Range, inches	**Tolerance (inch) 1/**
0.125 to 0.249	+ 0.040
0.250 to 0.374	+ 0.045
0.375 to 0.749	+0.055
0.750 to 0.999	+0.060
1.000 to 1.999	+0.070
2.000 to 2.999	+0.085
3.000 to 4.000	+0.100

1/ The entire range has a minus tolerance of 0.010 inch.

3.5.2 Straightness. Plates shall be of such straightness that the maximum edgewise curvature (depth of arc) shall not exceed 1/8 inch for any 5 ft. section along its length.

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3.5.3 Flatness. Flatness tolerances across the plate width shall conform to table IV.

TABLE IV. Flatness tolerances for plate.

Ordered Thickness (inch)		SPECIFIED WIDTH (INCH)								
		OVER	0.00	48.01	60.01	72.01	84.01	96.01	108.01	120.01	144.01
		THRU	48.00	60.00	72.00	84.00	96.00	108.00	120.00	144.00	
OVER	THRU	MAXIMUM PERMISSIBLE VARIATION IN FLATNESS - INCHES									
0.125	0.250		0.750	1.062	1.250	1.375	1.625	1.625	1.875	2.000	--
0.251	0.375		0.687	0.750	0.937	1.125	1.375	1.437	1.562	1.875	--
0.376	0.500		0.500	0.562	0.687	0.750	0.937	1.125	1.250	1.437	1.750
0.501	0.750		0.500	0.562	0.625	0.625	0.812	1.125	1.125	1.125	1.375
0.751	1.000		0.500	0.562	0.625	0.625	0.750	0.812	0.937	1.000	1.125
1.001	1.500		0.500	0.562	0.562	0.562	0.687	0.687	0.687	0.750	1.000
1.501	4.000		0.187	0.312	0.375	0.437	0.500	0.562	0.625	0.750	0.875

3.6 Marking for identification. Unless otherwise specified in the contract or order that MIL-STD-129 or any other markings shall be used (see 6.2), each plate shall be marked on one face with the manufacturer's name or trademark, the basic number and revision letter of this specification, the composition, the nominal plate thickness, and the lot number. Each plate shall be marked in lengthwise rows of characters recurring at intervals not greater than 3 feet, the rows being spaced not more than 6 inches apart and alternately staggered. The characters shall be not less than 3/8 inch in height and shall be applied using a suitable marking fluid whose residue shall contain no more than traces of halogen-bearing compounds and shall be capable of being removed in hot alkaline cleaning solution without rubbing. The markings shall have no deleterious effect on the plate material or its performance and shall be sufficiently stable to withstand normal handling.

3.6.1 Marking of ballistic test plates. Each ballistic test plate shall be marked with a plate identification code (see 3.6) and the letters ‘PRE’ if a first article test plate and ‘ACC’ if an acceptance test plate. When practicable, the marking shall be an impression stamped on the horizontal edge face of the ballistic test plates.

3.7 Heat treatment. Product shall be heat treated within prescribed temperature, time, and atmospheric condition ranges established during first article testing or commonly approved production runs.

3.7.1 Reheating. Unless otherwise approved by the procuring activity (see 6.2), after the final heat treatment, material shall not be reheated or processed above the final annealing, stress relieving, or aging temperatures or to a temperature that is 100° F below the beta transus, whichever is less. If plates are retreated above these limiting temperatures, they shall be so recorded and completely re-tested as a separate lot.

3.8 Ballistic test plate information. For each lot of titanium alloy armor plate a properly completed Titanium Armor Test Data Form (see figure 1) shall be submitted with each ballistic test plate that represents that particular processing lot.

3.9 Finish. Unless otherwise specified (see 6.2), plate shall have a surface finish in accordance with ASTM A480 after surface preparation (if necessary). Product shall be free of surface contamination by oxygen (alpha case), hydrogen, carbon, nitrogen, or other such harmful

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contaminants. The producer shall be permitted to remove minor surface imperfections by spot grinding if such grinding does not reduce the thickness of the material below the minimum permitted by the tolerance for the thickness ordered.

3.10 Workmanship. The material produced under this specification shall be uniform in quality and condition, and free from harmful alloy segregation and surface contamination by oxygen, nitrogen, or other contaminant and foreign material. The plate shall be clean, sound, smooth and free of defects detrimental to fabricability or serviceability such as hard spots, laminations, inclusions, pits, folds, seams, grind marks, and cracks.

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REQUEST FOR BALLISTIC TEST OF TITANIUM ARMOR

SPECIFICATION: MIL-DTL-46077	REVISION: H	AMENDMENT:
FIRING RECORD:	DATE:	TEST CONTRACT NO:

	PLATE MANUFACTURER / PRODUCER	PRIME CONTRACTOR:
Name:		
Address:		
POC:		
Phone No:		
Email:		

TEST ITEM IDENTIFICATION:	ORDERED THICKNESS:

Lot No. _____ Plate No. _____	Alloy: _____ Class: _____	Heat Treatment: _____ Finishing Temperature: _____

PURPOSE: ___ Acceptance ___ First Article ___ Development ___ Reduced Testing ___ Audit Testing

SAMPLE: ___ Primary ___ Retest (Firing Record No. of Failed Sample _____)

CHEMICAL ANALYSIS:

Al	V	Fe	O	C	N	H	Si			Ti
										Rem

MECHANICAL PROPERTIES:

UTS (psi):	0.2% YS (psi):	Elongation (%):

BALLISTIC TEST RESULTS:

	Obl. (deg)	Actual Thickness (in)	Required V_{50} (fps)	Actual V_{50} (fps)	Pass/ Fail
Test:					
Projectile:					
	NOTES:				

DO NOT FILL BELOW THIS LINE - TO BE COMPLETED BY GOVERNMENT OFFICIAL ONLY

LOT: [____ met] [____ failed to meet] the ballistic requirements of specification MIL-DTL-46077

Government Representative:	Date:	Supplier Representative:	Date:

FIGURE 1. Titanium Armor Test Data Form.

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4. VERIFICATION

4.1 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Conformance inspection (see 4.5).

4.2 Ballistic testing facility. Unless otherwise specified in the contract or purchase order (see 6.2), the ballistic test plates shall be forwarded to the Commander, USA ATC, ATTN: TEDT-AT-SL-V, Building 358, 6850 Lanyard Road, APG, MD 21005-5059 or to an approved Government facility for ballistic testing for first article or production acceptance.

4.3 Lot. A lot shall consist of all plates of the same declared chemistry, the same nominal thickness from the same heat, processed together as a unit. All plates in a lot shall receive the same final heat treatment within standard parameters, but not necessarily at the same time.

4.4 First article inspection. First article inspection, except as otherwise indicated in this specification, shall utilize the same requirements and test methods as the production acceptance inspection shown in 4.5.

4.5 Conformance inspection. Conformance inspection or production acceptance inspection shall include the examination of 4.7 and the tests of 4.8.

4.6 Sampling.

4.6.1 For first article inspection.

4.6.1.1 Chemical analysis. A sample for chemical analysis shall be taken from each heat.

4.6.1.2 Mechanical properties. At least two tension test specimens shall be taken from each lot. The location and type of specimens shall be as specified in 4.6.2.2.2 and 4.6.2.2.3.

4.6.1.3 Ballistic test samples. Two plates, of the thickness specified on the contract, shall be submitted to a Government approved facility (see 4.2) for ballistic tests in accordance with Appendix A. One sample shall be taken from the first plate fabricated and one from the last plate fabricated in the initial lot produced. When only one plate is fabricated, the samples shall be taken from opposite ends of the plate. The required ballistic testing plate dimensions and test projectiles for the ordered plate thickness ranges are defined in table V; all ballistic tests shall be conducted at zero degrees obliquity. First article approval may be obtained for each of the thickness ranges for each test projectile listed in table V by submission of two plates at one specified thickness within each thickness range. For example, a 1”-thick plate tested with the 20mm FSP can satisfy the range of thicknesses from 0.615” to 1.575” that all use the 20mm FSP as the test projectile.

4.6.2 For conformance inspection.

4.6.2.1 For chemical analysis. A sample sufficient to perform the analysis required to determine compliance with 3.2 shall be selected from each lot.

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4.6.2.2 For mechanical properties (See 3.3).

4.6.2.2.1 Number of samples. Two random samples shall be taken from each lot. If a lot consists of only one plate, only one sample shall be selected. At least one tension test specimen shall be made from each sample.

TABLE V. Ballistic test plate sizes and corresponding test projectiles for first article and acceptance testing.

Thickness Ranges (inches)	Minimum Size (inches)	Ballistic Test Projectile	Table for Ballistic Acceptance
0.125 to 0.250, excl	12 x 12	5.56mm M193	Table A-II
0.250 to 0.615, excl	12 x 12	Cal. .30 AP M2	Table A-III
0.615 to 1.575, excl	18 x 12	20-mm FSP	Table A-IV
1.575 to 1.950, excl	18 x 12	14.5-mm API B32	Table A-V
1.950 to 3.250, excl	18 x 12	20-mm API-T M602	Table A-VI
3.250 to 4.000	24 x 12	30-mm APDS	Table A-VII

4.6.2.2.2 Types of test specimens. Tension test specimens shall conform to the largest obtainable round cross-section specimen shown in ASTM E8/E8M.

4.6.2.2.3 Location of test specimens. Transverse test specimens shall be taken from the plate perpendicular to the primary rolling direction with the central longitudinal axis midway between the top and bottom surfaces of the plate. Longitudinal specimens shall be prepared in such a manner that the longitudinal axis of the specimen is parallel to the principal rolling direction of the sample with the central longitudinal axis midway between the top and bottom surfaces of the plate.

4.6.2.3 For ballistic testing. According to the thickness of the plate represented (ordered plate thickness) one plate, 12 inch by 12 inch for thicknesses less than 0.615 inch or 18 inch by 12 inch for thicknesses 0.615 inch to 3.249 inch or 24 inch by 12 inch for thicknesses 3.250 inch and greater shall be selected from every lot of material of the same thickness. In the event of reduced testing (see 4.8.4), multiple lots represented by a single acceptance ballistic test shall have been manufactured using the same heat treatment and other processing within standard mill parameters of a fixed process.

4.7 Examination.

4.7.1 Visual. Each piece in each lot shall be visually examined for compliance with the requirements for workmanship (see 3.9) and marking for identification (see 3.6).

4.7.2 Dimensions. All plates shall be subject to inspection for compliance with the dimensional and tolerance requirements (see 3.5).

4.8 Tests (first article and quality conformance).

4.8.1 Chemical composition. Conformance with the chemical composition requirements shall be determined using ASTM E539 X-ray emission spectrometric method for aluminum, vanadium and iron. Oxygen shall be analyzed by the inert gas fusion technique specified in ASTM E1409.

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Hydrogen shall be analyzed by the inert gas fusion thermal conductivity method specified in ASTM E1447. Carbon and nitrogen shall be determined using ASTM E2371. Other analytical methods may be used with the written approval of the acquiring activity. In case of a dispute, the results of chemical analysis by the ASTM E2371 method shall be used as the basis for acceptance or rejection.

4.8.2 Mechanical properties. Conformance to the mechanical property requirements shall be determined by testing at least two specimens (one longitudinal and one transverse) in the condition ordered from each sample selected in accordance with 4.6.2.2. Tension tests shall be accomplished in accordance with the following provisions.

4.8.2.1 Tension specimen configuration. The tension specimens shall conform to the following configuration in accordance with ASTM E8/E8M:

a. For product thickness up to 0.375 inch, standard "sheet-type" rectangular cross-section tension specimens shall be used.

b. For product thickness equal to or greater than 0.375 inch, standard round cross-section tension specimens with a 0.250 inch diameter minimum reduced test section shall be used.

4.8.2.2 Tension specimen preparation. Unless the lot consists of only one unit, not more than one specimen per grain direction per test condition shall be taken from each sample. Longitudinal specimens shall be prepared in such a manner that the longitudinal axis of the specimen is parallel to the principal rolling direction of the sample. Transverse specimens shall be prepared in such a manner that the longitudinal axis of the specimen is perpendicular to the principal rolling direction and the thickness direction of the sample.

4.8.2.3 Tension test method. Tension tests shall be accomplished in accordance with ASTM E8/E8M, except that the strain rate shall be limited to 0.003 to 0.007 inch/inch per minute for the measurement of the yield strength: after that it is to be increased so as to produce tensile failure in approximately one additional minute. In case of dispute, the results of referee tension tests performed on a tensile machine having a strain pacer and using a strain rate of 0.005 inch/inch per minute through the yield strength and a minimum crosshead speed of 0.10 inch per minute when the yield strength has been exceeded shall govern.

4.8.3 Ballistic tests. Plate thickness as determined by the ballistic test agency shall be used to determine the required protection ballistic limit for that plate and shall be taken as the average of 4 thickness measurements read to the nearest 0.001 inch and the average thickness reported to the nearest 0.005 inch. At least one measurement shall be taken near each edge of the plate at a distance of at least 1 inch from the edge. The V_{50} ballistic tests shall be performed in accordance with Appendix A.

4.8.3.1 Cracking. No through cracks greater than 2.5 inch in length on any plate surface are acceptable after ballistic testing.

4.8.4 Reduced testing for acceptance. Upon the approval of the procuring activity (see 6.2), reduction in the amount of testing per ASQ Z1.4 may be authorized provided the results on consecutive lots indicate that a uniform product meeting the requirements is being produced. Reduction in testing shall only be considered if steady production exists as determined by the

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procuring activity. Testing for a given plate thickness shall return to standard (non-reduced testing) conditions of one plate per lot, whenever a ballistic test plate fails to meet ballistic requirements.

4.8.4.1 First level reduced testing. Testing frequency may be reduced, upon approval by a responsible authority, to one ballistic test plate for every three lots of material of the same thickness providing the preceding five ballistic test plates for that thickness have met ballistic requirements.

4.8.4.2 Second level reduced testing. Testing frequency may be further reduced, upon approval by a responsible authority, to one ballistic test plate for every five lots of material of the same thickness providing the preceding ten ballistic test plates of that thickness have met ballistic requirements.

4.9 Rejection.

4.9.1 First article. Failure of the first article samples to meet any of the testing and inspection requirements of this specification indicates failure of the process to produce acceptable product.

4.9.2 Retests. Resubmission of first article samples shall not be made until the contractor has demonstrated that he has made necessary corrections in processing the material to the satisfaction of the procuring activity.

4.9.3 Quality conformance.

4.9.3.1 Examination. A lot shall be subject to rejection for failure to meet the visual and dimensional requirements of 4.7.

4.9.3.2 Tests. A lot shall be subject to rejection for failure to meet any of the test requirements of 4.8.

4.9.3.3 Retests. At the discretion of the producer, a rejected lot may be retested as specified below.

4.9.3.3.1 Non-ballistic retests. All failed sample units from the original sample shall be excluded from the retest lot. Four new sample units shall be taken and tested in accordance with 4.8.2, as applicable. Failure of any retest specimen to meet the applicable property requirements shall be cause for rejection of the entire lot and no further retesting shall be permitted.

4.9.3.3.2 Non-ballistic retest lot. If the retest lot is accepted, the producer may, at his discretion, retest each sample unit which shows failure at one end only. Units which show failure at both ends shall not be retested and shall be rejected. Each sample unit acceptable for retesting of mechanical properties shall be retested as follows: Two test specimens shall be taken from the failed end in either or both directions, longitudinal and transverse, which showed unacceptable mechanical properties, and these specimens shall be tested in accordance with 4.8.2. If any retest shows that the mechanical properties do not conform to the requirements of table II then the parent unit may be cut back at the failed end and further retested until either: (a) both ends show acceptable properties, or (b) the length of the material remaining in the parent unit is less than the minimum specified length in the acquisition documents, whichever comes first. Acceptance or rejection of the parent unit shall then be made as appropriate.

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4.9.3.3.3 Ballistic retests. In the event of failure of any one or all of the ballistic test plates, the failed lot(s) may be reprocessed and two plates submitted for retest from each reprocessed lot. If a failure then occurs, the lot shall be permanently rejected.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel components is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The armor specified herein is intended for use on combat vehicles. Class 1 is equivalent to material specified by previous revisions of MIL-A-46077, which is an Extra Low Interstitial (ELI) grade. Class 2 is equivalent to material specified by AB-1 composition originally defined under the cancelled MIL-T-9046J and SAE AMS-T-9046 specifications and still referenced within numerous engineering drawings and contracts. These specifications were replaced by three SAE AMS specifications based upon the following heat treated conditions:

Annealed	AMS 4911
Solution Treated	AMS 4903
Solution treated and aged (STA)	AMS 4904

Class 3 permits a higher oxygen level which would be suitable for applications where maximum ductility and toughness are not necessary, such as appliqué. Class 4 permits compositions different from the conventional Ti-6Al-4V alloy and should be used as appliqué armor only. A weldability study should be conducted prior to any welding of Class 4 armor.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Product form size (thickness, width, and length) and quantity.

c. Specify classification, composition, and heat treat condition (annealed) (see 1.2).

d. When first article sample is required (see 3.1).

e. If markings for identification are specified (see 3.6).

f. If reheating is allowed (see 3.7).

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g. If surface finish is specified (see 3.9).

h. Name of inspection agency when inspection should be performed by other than the contractor (see 4.1).

i. If contractor's facility is not allowed for performing inspection (see 4.1).

j. If another Government ballistic testing facility is selected by the procuring activity (see 4.2).

k. If reduced testing is allowed (see 4.8.4).

l. Packaging requirements (see 5.1).

6.3 Marking. Suggested wording to be included in the contract or purchase order "Marking for shipment and storage should be in accordance with MIL-STD-129".

6.4 Stress corrosion. Certain processing procedures and service conditions may cause these titanium alloy products to be subject to stress corrosion cracking. SAE ARP 982 recommends practices to minimize the susceptibility to such stress corrosion cracking.

6.5 Annealing Temperature. It has been found that an annealing temperature of 1650°F provides optimum ballistic properties for class 1 material. Reference: *Effect of Annealing Temperature on the Ballistic Limit Velocity of Ti-6Al-4V ELI,* Matthew Burkins and William Love, Proceedings of the 16th International Symposium on Ballistics, San Francisco, CA 23-27 September 1996.

6.6 Definitions.

6.6.1 Plate. A flat rolled product of 0.125 inch or over in thickness and over 12 inches in length with the width at least five times the thickness.

6.6.2 Capability. The words "shall be capable of" are used herein to indicate characteristics or properties required in the product but for which testing of each lot is not required. However, if such testing is performed, products not conforming to the requirements of this specification should be subject to rejection.

6.6.3 Weldable. For the purposes of this specification, a plate is defined as "weldable" if it has a composition within the limits of its class AWS 1.9 (Ti welding spec).

6.7 Relationship of mechanical properties to ballistic requirements. The minimum mechanical properties specified may not assure titanium armor plate meeting the specified ballistic requirements.

6.8 Plate tolerances. Titanium alloy plate is normally specified to tolerances established for corrosion-resisting steel plate as specified in the applicable sections of ASTM A480 for thickness, width, length, camber, and flatness.

6.9 Density. The density of the specified classes listed in this specification are as follows:

Class 1 (Ti-6Al-4V) = 0.16 lb/in^3 (4.43 g/cm^3).

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Class 2 (Ti-6Al-4V) = 0.16 lb/in³ (4.43 g/cm³).

Class 3 (Ti-6Al-4V) = 0.16 lb/in³ (4.43 g/cm³).

Class 4 (range) = 0.158 lb/in³ through 0.165 lb/in³ (4.373 g/cm³ through 4.567 g/cm³).

Class 4a (ATI 425) = 0.161 lb/in³ (4.456 g/cm³).

Class 4b (Timet 108) = 0.161 lb/in³ (4.456 g/cm³).

6.10 Metric units. When metric divisions are required, units for inch, foot, foot-pounds, feet per second, pounds per square inch, and Fahrenheit may be converted to the metric equivalent by multiplying them by the following conversion factors:

English	Multiply by	Equals	Metric SI unit
inch	0.0254	=	meter (m)
foot	0.3048	=	meter (m)
pound	0.4536	=	kilogram (kg)
foot-lb	1.3558	=	joule (j)
feet/second	0.3048	=	meter per second (m/s)
pounds/sq. inch	0.00689	=	Mega Pascal (MPa)
pound/cubic inch	27.6799	=	gram (g) per cubic centimeter (cm)
Fahrenheit (°F)	(t°F - 32)/1.8 = t°C		Celsius (°C)

6.11 Subject term (key word) listing.

Ballistic limit
Caliber .30 AP
5.56-mm M193
Extra Low Interstitial (ELI) grade
Stress Corrosion Cracking
Vehicles
14.5-mm API
20-mm API-T M602
20-mm FSP
30-mm APDS

6.12 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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APPENDIX A

BALLISTIC TESTING OF ARMOR PLATE, TITANIUM ALLOY, WELDABLE

A.1 SCOPE

A.1.1 Scope. This appendix covers the minimum ballistic limits for acceptable requirements of titanium alloy armor plate when tested in accordance with the provisions of this specification. When there is mutual agreement between contractor and procuring activity, this appendix becomes a mandatory part of this specification and the information contained herein is intended for compliance.

A.2 APPLICABLE DOCUMENTS

A.2.1 Government documents.

A.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the contract or purchase order.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-662 - V_{50} Ballistic Test for Armor

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

A.3 DEFINITIONS

A.3.1 Complete penetration, (CP). A complete penetration occurs when the impacting projectile, or any fragment thereof, or any fragment of the test specimen perforates the witness plate, resulting in a crack or hole which permits light passage when a 60-watt, 110-volt bulb is placed proximate to the witness plate.

A.3.2 Fair impact. An impact may be considered fair when an un-yawed fragment simulator or test projectile strikes an unsupported area of the target material at a specified obliquity and separated from another impact or the edge of the plate, hole, crack, or spalled area by an undisturbed area of at least one caliber.

A.3.3 Gap. A gap is the difference in fps between the high partial penetration velocity and the low complete penetration velocity used to compute the ballistic limit when the high partial penetration velocity is lower than the low complete penetration velocity.

A.3.4 Partial penetration, (PP). Any fair impact which is not a complete penetration may be considered a partial penetration.

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A.3.5 Witness plate. A witness plate is normally a 0.014 inch thick sheet of 5052-H36 aluminum alloy or a 0.020 inch thick sheet of 2024-T3 aluminum alloy placed 6 inches (+ 1/2 inch) behind and parallel to the test plates or other ballistic sample.

A.3.6 Spalling. The detachment or delamination of a layer of material in the area surrounding the location of impact, which may occur on either the front or rear surfaces of the armor. Ejected spall may impact the witness plate and cause a CP even when the projectile has not exited the plate.

A.4 REQUIREMENTS

A.4.1 Resistance to penetration. The minimum required V_{50} ballistic limit shall be in accordance with the values shown in tables A-II through A-VII.

A.4.2 V_{50} protection ballistic limit, BL(P). A BL(P) consists of an equal number of complete and partial penetrations attained by the up- and down-firing method. All BL(P)s should be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Table A-I delineates the order of priority.

Table A-I. Ballistic limits (protection criteria).

No. of Rounds in BL(P)	Maximum Allowable Velocity Spread (fps)	Maximum No. of Rounds to Be Used
4	60	12 1/
6	90	12
10	90 2/	as required

1/ Firing should continue until either a 4-round or 6-round BL(P) has been attained, whichever comes first in the firing order. If these occur simultaneously the 6-round BL(P) should be reported. If, after 12 rounds have been fired and neither a 4 or 6 round BL(P) has been determined, then firing should continue until a 10-round BL(P), having a maximum velocity spread of 125 fps, has been determined.

2/ In the event that a high partial penetration velocity occurs which is more than 90 fps above the low complete penetration velocity, then the ballistic limit should consist of the 5 highest partial penetration velocities and the 5 lowest complete penetration velocities. The maximum velocity spread should be kept as small as possible without deviating from the normal up-and-down method of firing.

A.4.2.1 Recomputed BL(P). In the event that after following the above procedures the BL(P) is within ± 10 fps of the minimum allowable ballistic limit and a gap (high partial penetration velocity less than the low complete penetration velocity) exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less (see A.5.1.2.3). The BL(P) should then be recomputed as before using the highest partial penetration velocities and the lowest complete penetration velocities. The recomputed BL(P) should be the one reported on the firing record.

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APPENDIX A

A.5 TESTS

A.5.1 Ballistic tests. Testing shall be in accordance with MIL-STD-662, V_{50} Ballistic Test for Armor, except that nothing in this procedure shall be construed to supersede or invalidate the requirements of this specification.

A.5.1.1 Temperature Conditioning. Prior to the test, the test item(s) shall be temperature conditioned at least eight hours. Thermostatic control shall be such that the average temperature of the item during the test shall be 72 ± 15°F (22 ± 8°C).

A.5.1.2 Protection ballistic limit, BL(P).

A.5.1.2.1 Normal circumstances. The BL(P) shall consist of an equal number of fair impact complete and partial penetration velocities attained by the up-and-down firing method. All BL(P)'s shall be computed using the highest partial penetration velocities and the lowest complete penetration velocities. Firing shall continue until either a 4-round BL(P) having a maximum velocity spread of 60 fps or a 6-round BL(P) having a maximum velocity spread of 90 fps has been attained, whichever comes first in the normal sequence of firing. If both occur simultaneously, the 6-round BL(P) shall be reported.

A.5.1.2.2 Large zone of mixed results. In the event that the zone of mixed results (difference between the high partial penetration velocity and the low complete penetration velocity, the PP velocity being higher than the low CP velocity) exceeds 90 fps, the firing data shall be compared with the specification minimum ballistic requirements. If the lowest complete penetration velocity is equal to or above the minimum specified ballistic limit velocity for the plate thickness, the ballistic limit shall be computed on the basis of 4- or 6-rounds using the smallest possible velocity spread. If the lowest complete penetration velocity is below the minimum allowable ballistic limit velocity, then testing shall continue until a 10-round ballistic limit has been attained using the smallest possible velocity spread. Ten-round ballistic limits shall be reported as agreed upon between the contractor and procuring activity.

A.5.1.2.3 Reduction of large velocity gap in borderline cases. If the ballistic limit, which has been determined, is within ± 10 fps from the minimum allowable ballistic limit and a gap exists which is greater than 25 fps, then another round, or rounds, shall be fired to reduce the gap to 25 fps or less. The ballistic limit shall then be recomputed using the above criteria. The recomputed BL(P) shall be reported as the BL(P) of the plate (in borderline cases, a reduction of the gap between the high partial penetration velocity and the low complete velocity should result in a better evaluation of the BL(P)).

A.5.2 Rejection and retest of ballistic plates.

A.5.2.1 First article tests (rejection). Unless otherwise specified in the contract or purchase order, failure of any of the first article test plates to meet the minimum ballistic requirements shown in the appendix of this specification indicates failure of the product and process.

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A.5.2.2 First article (retests). Resubmission of ballistic retest plates shall not be made until the manufacturer has made the necessary corrections in the processing of the material to the satisfaction of the procuring activity. Two retest plates shall be submitted for first article testing, and both tests shall pass; otherwise, the armor material shall be rejected.

A.5.2.3 Acceptance tests (rejection). Unless otherwise specified in the contract or purchase order, failure of a test plate to meet the ballistic requirements indicates failure of the lot; however, the final decision shall depend on the outcome of retests, if submitted.

A.5.2.4 Acceptance tests (retests). If a test plate representing a lot fails to meet the ballistic requirement, the manufacturer, upon notification of the failure may submit at his expense two additional test plates from the same lot for ballistic retest. If either of these plates fails the ballistic test, the lot shall be rejected. The manufacturer may elect to resubmit the lot after retreatment of the entire lot by submitting two additional test plates. If either of these plates fails, the lot shall be permanently rejected.

A.5.3 Disposal of ballistic test plates.

A.5.3.1 First article test plates. Upon request of the applicant within 15 days after ballistic testing, first article plates shall be returned "as is" to the applicant, at his expense, unless the plates were destroyed in testing.

A.5.3.2 Acceptance test plates. Acceptance test plates that comply with the requirements of this specification are considered part of the lot they represent, and ownership of the test plates passes to the Government with the acceptance of that lot. Acceptance test plates that fail to comply with the requirements of this specification are considered part of the lot they represent and remain the property of the producer just as does the rejectable lot they represent. The failed plates shall be returned, upon request, as in A.5.3.1.

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APPENDIX A

TABLE A-II. Minimum required ballistic limits - 5.56mm M193 at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.110	1752	0.170	2078	0.230	2318
0.115	1781	0.175	2102	0.235	2335
0.120	1810	0.180	2125	0.240	2352
0.125 1/	1839	0.185	2147	0.245	2368
0.130	1867	0.190	2169	0.249 2/	2380
0.135	1894	0.195	2190	0.250	2383
0.140	1921	0.200	2211	0.255	2399
0.145	1947	0.205	2231	0.260	2413
0.150	1973	0.210	2251	0.265	2427
0.155	1998	0.215	2270	0.270	2441
0.160	2023	0.220	2289	0.275	2454
0.165	2048	0.225	2308	0.280	2467

1/ Specification requirements begin with this ordered thickness.
2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-III. Minimum required ballistic limits - caliber .30 AP M2 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.240	1116	0.385	1715	0.530	2158
0.245	1142	0.390	1732	0.535	2171
0.250 1/	1167	0.395	1749	0.540	2185
0.255	1192	0.400	1766	0.545	2199
0.260	1216	0.405	1783	0.550	2212
0.265	1239	0.410	1799	0.555	2225
0.270	1263	0.415	1815	0.560	2239
0.275	1286	0.420	1832	0.565	2252
0.280	1308	0.425	1848	0.570	2265
0.285	1330	0.430	1864	0.575	2278
0.290	1352	0.435	1879	0.580	2291
0.295	1373	0.440	1895	0.585	2304
0.300	1394	0.445	1910	0.590	2317
0.305	1415	0.450	1926	0.595	2330
0.310	1436	0.455	1941	0.600	2342
0.315	1456	0.460	1956	0.605	2355
0.320	1476	0.465	1971	0.610	2368
0.325	1496	0.470	1986	0.614 2/	2378
0.330	1515	0.475	2001	0.615	2380
0.335	1534	0.480	2016	0.620	2393
0.340	1553	0.485	2031	0.625	2405
0.345	1572	0.490	2045	0.630	2417
0.350	1591	0.495	2059	0.635	2430
0.355	1609	0.500	2074	0.640	2442
0.360	1627	0.505	2088	0.645	2454
0.365	1645	0.510	2102	0.650	2466
0.370	1663	0.515	2116	0.655	2478
0.375	1681	0.520	2130	0.660	2490
0.380	1698	0.525	2144	0.665	2502

1/ Specification requirements begin with this ordered thickness.
2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
0.595	1866	0.775	2324	0.955	2963
0.600	1875	0.780	2341	0.960	2981
0.605	1885	0.785	2357	0.965	2999
0.610	1894	0.790	2374	0.970	3017
0.615 1/	1904	0.795	2391	0.975	3035
0.620	1913	0.800	2408	0.980	3053
0.625	1922	0.805	2425	0.985	3071
0.630	1932	0.810	2442	0.990	3089
0.635	1942	0.815	2459	0.995	3107
0.640	1952	0.820	2477	1.000	3124
0.645	1963	0.825	2494	1.005	3142
0.650	1973	0.830	2512	1.010	3160
0.655	1985	0.835	2529	1.015	3177
0.660	1996	0.840	2547	1.020	3195
0.665	2008	0.845	2565	1.025	3212
0.670	2020	0.850	2582	1.030	3230
0.675	2032	0.855	2600	1.035	3247
0.680	2045	0.860	2618	1.040	3264
0.685	2057	0.865	2636	1.045	3281
0.690	2070	0.870	2654	1.050	3299
0.695	2084	0.875	2672	1.055	3316
0.700	2097	0.880	2690	1.060	3332
0.705	2111	0.885	2708	1.065	3349
0.710	2125	0.890	2727	1.070	3366
0.715	2139	0.895	2745	1.075	3383
0.720	2154	0.900	2763	1.080	3399
0.725	2168	0.905	2781	1.085	3416
0.730	2183	0.910	2799	1.090	3432
0.735	2198	0.915	2817	1.095	3448
0.740	2213	0.920	2836	1.100	3465
0.745	2229	0.925	2854	1.105	3481
0.750	2244	0.930	2872	1.110	3497
0.755	2260	0.935	2890	1.115	3513
0.760	2276	0.940	2908	1.120	3528
0.765	2292	0.945	2927	1.125	3544
0.770	2308	0.950	2945	1.130	3560

1/ Specification requirements begin with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-IV. Minimum required ballistic limits - 20mm fragment simulating projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.135	3575	1.310	4046	1.485	4395
1.140	3591	1.315	4057	1.490	4404
1.145	3606	1.320	4068	1.495	4413
1.150	3621	1.325	4080	1.500	4422
1.155	3636	1.330	4091	1.505	4431
1.160	3651	1.335	4102	1.510	4440
1.165	3666	1.340	4113	1.515	4449
1.170	3681	1.345	4123	1.520	4458
1.175	3695	1.350	4134	1.525	4466
1.180	3710	1.355	4145	1.530	4475
1.185	3724	1.360	4155	1.535	4484
1.190	3738	1.365	4166	1.540	4493
1.195	3752	1.370	4176	1.545	4502
1.200	3766	1.375	4186	1.550	4511
1.205	3780	1.380	4196	1.555	4520
1.210	3794	1.385	4207	1.560	4529
1.215	3808	1.390	4217	1.565	4538
1.220	3821	1.395	4227	1.570	4547
1.225	3835	1.400	4236	1.574 2/	4554
1.230	3848	1.405	4246	1.575	4556
1.235	3861	1.410	4256	1.580	4565
1.240	3874	1.415	4266	1.585	4574
1.245	3887	1.420	4275	1.590	4583
1.250	3900	1.425	4285	1.595	4592
1.255	3913	1.430	4294	1.600	4601
1.260	3926	1.435	4304	1.605	4610
1.265	3938	1.440	4313	1.610	4619
1.270	3950	1.445	4322	1.615	4628
1.275	3963	1.450	4331	1.620	4637
1.280	3975	1.455	4341	1.625	4646
1.285	3987	1.460	4350	1.630	4655
1.290	3999	1.465	4359	1.635	4664
1.295	4011	1.470	4368	1.640	4673
1.300	4022	1.475	4377	1.645	4682
1.305	4034	1.480	4386	1.650	4691

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-V. Minimum required ballistic limits - 14.5-mm API B32 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.565	2679	1.725	2871	1.885	3052
1.570	2685	1.730	2877	1.890	3057
1.575 1/	2691	1.735	2883	1.895	3063
1.580	2697	1.740	2889	1.900	3068
1.585	2704	1.745	2895	1.905	3074
1.590	2710	1.750	2900	1.910	3079
1.595	2716	1.755	2906	1.915	3085
1.600	2722	1.760	2912	1.920	3090
1.605	2728	1.765	2918	1.925	3096
1.610	2734	1.770	2923	1.930	3101
1.615	2740	1.775	2929	1.935	3106
1.620	2746	1.780	2935	1.940	3112
1.625	2753	1.785	2940	1.945	3117
1.630	2759	1.790	2946	1.949 2/	3121
1.635	2765	1.795	2952	1.950	3122
1.640	2771	1.800	2957	1.955	3128
1.645	2777	1.805	2963	1.960	3133
1.650	2783	1.810	2969	1.965	3138
1.655	2789	1.815	2974	1.970	3144
1.660	2795	1.820	2980	1.975	3149
1.665	2801	1.825	2986	1.980	3154
1.670	2807	1.830	2991	1.985	3160
1.675	2813	1.835	2997	1.990	3165
1.680	2818	1.840	3002	1.995	3170
1.685	2824	1.845	3008	2.000	3176
1.690	2830	1.850	3013	2.005	3181
1.695	2836	1.855	3019	2.010	3186
1.700	2842	1.860	3024	2.015	3191
1.705	2848	1.865	3030	2.020	3197
1.710	2854	1.870	3036	2.025	3203
1.715	2860	1.875	3041	2.030	3208
1.720	2866	1.880	3047	2.035	3213

1/ Specification requirements begin with this ordered thickness.
2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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TABLE A-VI. Minimum required ballistic limits - 20-mm API-T M602 projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
1.935	2330	2.090	2471	2.245	2611
1.940	2334	2.095	2475	2.250	2616
1.945	2339	2.100	2480	2.255	2620
1.950 1/	2343	2.105	2485	2.260	2625
1.955	2348	2.110	2490	2.265	2629
1.960	2352	2.115	2494	2.270	2634
1.965	2357	2.120	2499	2.275	2638
1.970	2361	2.125	2503	2.280	2643
1.975	2365	2.130	2508	2.285	2647
1.980	2370	2.135	2512	2.290	2652
1.985	2375	2.140	2517	2.295	2656
1.990	2380	2.145	2521	2.300	2661
1.995	2384	2.150	2526	2.305	2665
2.000	2389	2.155	2530	2.310	2670
2.005	2393	2.160	2535	2.315	2674
2.010	2398	2.165	2539	2.320	2679
2.015	2402	2.170	2544	2.325	2683
2.020	2407	2.175	2548	2.330	2687
2.025	2411	2.180	2553	2.335	2691
2.030	2416	2.185	2557	2.340	2696
2.035	2421	2.190	2562	2.345	2700
2.040	2426	2.195	2566	2.350	2705
2.045	2430	2.200	2571	2.355	2709
2.050	2435	2.205	2575	2.360	2714
2.055	2439	2.210	2580	2.365	2718
2.060	2444	2.215	2584	2.370	2723
2.065	2448	2.220	2589	2.375	2727
2.070	2453	2.225	2593	2.380	2732
2.075	2457	2.230	2598	2.385	2736
2.080	2462	2.235	2602	2.390	2740
2.085	2466	2.240	2607	2.395	2744

1/ Specification requirements begin with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 20-mm API-T M602 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.400	2749	2.560	2889	2.720	3026
2.405	2753	2.565	2893	2.725	3030
2.410	2758	2.570	2898	2.730	3034
2.415	2762	2.575	2902	2.735	3038
2.420	2767	2.580	2906	2.740	3043
2.425	2771	2.585	2910	2.745	3047
2.430	2776	2.590	2915	2.750	3051
2.435	2780	2.595	2919	2.755	3055
2.440	2784	2.600	2923	2.760	3060
2.445	2788	2.605	2927	2.765	3064
2.450	2793	2.610	2932	2.770	3068
2.455	2797	2.615	2936	2.775	3072
2.460	2802	2.620	2941	2.780	3077
2.465	2806	2.625	2945	2.785	3081
2.470	2811	2.630	2949	2.790	3085
2.475	2815	2.635	2953	2.795	3089
2.480	2819	2.640	2958	2.800	3093
2.485	2823	2.645	2962	2.805	3097
2.490	2828	2.650	2966	2.810	3102
2.495	2832	2.655	2970	2.815	3106
2.500	2837	2.660	2975	2.820	3110
2.505	2841	2.665	2979	2.825	3114
2.510	2846	2.670	2983	2.830	3119
2.515	2850	2.675	2987	2.835	3123
2.520	2854	2.680	2992	2.840	3127
2.525	2858	2.685	2996	2.845	3131
2.530	2863	2.690	3000	2.850	3135
2.535	2867	2.695	3004	2.855	3139
2.540	2872	2.700	3009	2.860	3144
2.545	2876	2.705	3013	2.865	3148
2.550	2880	2.710	3017	2.870	3152
2.555	2884	2.715	3021	2.875	3156

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VI. Minimum required ballistic limits - 20-mm API-T M602 projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
2.880	3160	3.035	3288	3.190	3413
2.885	3164	3.040	3292	3.195	3417
2.890	3169	3.045	3296	3.200	3421
2.895	3173	3.050	3300	3.205	3425
2.900	3177	3.055	3304	3.210	3429
2.905	3181	3.060	3308	3.215	3433
2.910	3185	3.065	3312	3.220	3437
2.915	3189	3.070	3316	3.225	3441
2.920	3193	3.075	3320	3.230	3445
2.925	3197	3.080	3325	3.235	3449
2.930	3202	3.085	3329	3.240	3453
2.935	3206	3.090	3333	3.245	3457
2.940	3210	3.095	3337	3.249 2/	3460
2.945	3214	3.100	3341	3.250	3461
2.950	3218	3.105	3345	3.255	3465
2.955	3222	3.110	3349	3.260	3469
2.960	3226	3.115	3353	3.265	3473
2.965	3230	3.120	3357	3.270	3477
2.970	3235	3.125	3361	3.275	3481
2.975	3239	3.130	3365	3.280	3485
2.980	3243	3.135	3369	3.285	3489
2.985	3247	3.140	3373	3.290	3493
2.990	3251	3.145	3377	3.295	3497
2.995	3255	3.150	3381	3.300	3501
3.000	3259	3.155	3385	3.305	3505
3.005	3263	3.160	3389	3.310	3509
3.010	3267	3.165	3393	3.315	3513
3.015	3271	3.170	3397	3.320	3517
3.020	3276	3.175	3401	3.325	3521
3.025	3280	3.180	3405	3.330	3525
3.030	3284	3.185	3409	3.335	3529

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VII. Minimum required ballistic limits - 30-mm APDS projectiles at 0° obliquity.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.235	3491	3.380	3606	3.525	3722
3.240	3495	3.385	3610	3.530	3726
3.245	3499	3.390	3614	3.535	3730
3.250 1/	3503	3.395	3618	3.540	3734
3.255	3507	3.400	3622	3.545	3738
3.260	3511	3.405	3626	3.550	3742
3.265	3514	3.410	3630	3.555	3746
3.270	3518	3.415	3634	3.560	3750
3.275	3522	3.420	3638	3.565	3754
3.280	3526	3.425	3642	3.570	3758
3.285	3530	3.430	3646	3.575	3762
3.290	3534	3.435	3650	3.580	3766
3.295	3538	3.440	3654	3.585	3770
3.300	3542	3.445	3658	3.590	3774
3.305	3546	3.450	3662	3.595	3778
3.310	3550	3.455	3666	3.600	3782
3.315	3554	3.460	3670	3.605	3786
3.320	3558	3.465	3674	3.610	3790
3.325	3562	3.470	3678	3.615	3794
3.330	3566	3.475	3682	3.620	3798
3.335	3570	3.480	3686	3.625	3802
3.340	3574	3.485	3690	3.630	3806
3.345	3578	3.490	3694	3.635	3810
3.350	3582	3.495	3698	3.640	3814
3.355	3586	3.500	3702	3.645	3818
3.360	3590	3.505	3706	3.650	3822
3.365	3594	3.510	3710	3.655	3826
3.370	3598	3.515	3714	3.660	3830
3.375	3602	3.520	3718	3.665	3834

1/ Specification requirements begin with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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APPENDIX A

TABLE A-VII. Minimum required ballistic limits - 30-mm APDS projectiles at 0° obliquity - Continued.

Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps	Thickness, inches	Required BL(P), fps
3.670	3838	3.820	3957	3.970	4077
3.675	3842	3.825	3961	3.975	4081
3.680	3846	3.830	3965	3.980	4085
3.685	3850	3.835	3969	3.985	4089
3.690	3854	3.840	3973	3.990	4093
3.695	3858	3.845	3977	3.995	4097
3.700	3861	3.850	3981	4.000 2/	4101
3.705	3865	3.855	3985	4.005	4105
3.710	3869	3.860	3989	4.010	4109
3.715	3873	3.865	3993	4.015	4113
3.720	3877	3.870	3997	4.020	4117
3.725	3881	3.875	4001	4.025	4121
3.730	3885	3.880	4005	4.030	4125
3.735	3889	3.885	4009	4.035	4129
3.740	3893	3.890	4013	4.040	4133
3.745	3897	3.895	4017	4.045	4137
3.750	3901	3.900	4021	4.050	4141
3.755	3905	3.905	4025	4.055	4145
3.760	3909	3.910	4029	4.060	4149
3.765	3913	3.915	4033	4.065	4153
3.770	3917	3.920	4037	4.070	4157
3.775	3921	3.925	4041	4.075	4161
3.780	3925	3.930	4045	4.080	4165
3.785	3929	3.935	4049	4.085	4169
3.790	3933	3.940	4053	4.090	4173
3.795	3937	3.945	4057	4.095	4177
3.800	3941	3.950	4061	4.100	4181
3.805	3945	3.955	4065	4.105	4185
3.810	3949	3.960	4069	4.110	4189
3.815	3953	3.965	4073	4.115	4193

2/ Specification requirements end with this ordered thickness.

Note: Tabulated values on either side of the ordered thicknesses are for interpolation of BL(P) requirements on undersize or oversize plates.

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CONCLUDING MATERIAL

Custodians:
Army - MR
Navy - OS
Air Force - 11

Review activities:
Army - AT, AV, TE, MI
Navy - AS
DLA - CQ

Preparing activity:
Army - MR

(Project 9535-2022-007)

Agent:
DLA - GS

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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